Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/02/2006
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Project Title:	Induction Of Bystander Effects By High LET Radiation In Cells And In Tissue Models		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
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Zip Code:	02114-2696	**Congressional District:**	9
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Contact Monitor:		**Contact Phone:**	
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With charged particles such as alpha-particles and low energy protons, significant differences from direct DNA damage-mediated responses have been observed in a number of biological systems. Occurrence of these non-targeted effects, including bystander effects and genomic instability, calls into question the use of a simple linear extrapolation of cancer risks from high dose data to low doses. Two important considerations are of particular relevance to NASA. First, the cancer risks from high doses of high LET radiations such as heavy ions and of energetic protons are not well understood, making it difficult to extrapolate to low doses. Second, the occurrence of bystander effects in cells exposed to heavy ions or energetic protons, as found in space, has not been evaluated to date. The overall goals of this project are to test the hypothesis that heavy ions and energetic protons, as encountered in space, efficiently cause bystander effects in neighboring cells not directly hit by the radiation and to study the mechanisms involved. Hence, the studies specifically address the Research Emphasis for FY2003 of “Molecular radiation biology of carcinogenesis”, as described

Task Description:	in the NASA Research Announcement. The studies will aid in assessing the risks to astronauts of carcinogenesis as a result of exposures to radiation in space. There are four aims. 1. Determine induction of bystander effect(s) by heavy ions compared to X-rays, protons (low and high energy) and alpha particles. Cultured human diploid fibroblasts and keratinocytes will be used with endpoints specific for each cell type and including production of micronuclei, apoptosis, clonogenicity and gene expression. Dose response curves for the induction of bystander effects will be obtained for a range of particles and energies available at the Brookhaven National Laboratory (BNL) NASA Space Radiation Laboratory (NSRL). Results with heavy particles and high energy protons will be compared with those obtained using full field irradiations with X-rays, protons and alpha particles. We will test the hypothesis that heavy ions and high energy protons effectively induce bystander effects in neighboring cells not traversed by the particles. 2. Test the hypothesis that heavy ion-induced bystander effect(s) can be modified by exposure of cells to reactive oxygen species (ROS), an overlying oxidative stress. Cells will be exposed to acute and chronic ROS, before, during and/or after exposure to the heavy ions, and bystander effect(s) will be evaluated. 3. Test the hypothesis that exposure to low LET radiations alters the heavy ion-induced bystander effect(s). Cells will be exposed to very low doses of low LET radiation, before and/or after exposure to the heavy ions, and bystander effect(s) will be evaluated. 4. Begin development of 3-D tissue models for assessment of bystander effects from heavy ions and energetic protons. A long-term goal of this work is to investigate the role of bystander effects in tissues, composed of mixed cell types in 3-D arrangements. These studies will test the hypothesis that bystander effects are also seen in tissues. The overall goals of this project are to test the hypothesis that heavy ions and energetic protons, as encountered in space, efficiently cause bystander effects in neighboring cells not directly hit by the radiation and to study the mechanisms involved. The studies specifically address the Research Emphasis for FY2003 of “Molecular radiation biology of carcinogenesis”, as described in the NASA Research Announcement. The studies will aid in assessing the risks to astronauts of carcinogenesis as a result of exposures to radiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research should provide added insight into mechanisms involved in radiation-induced carcinogenesis and cell killing. Such new insight may be useful for developing new approaches for prevention or treatment of cancer.
Task Progress:	Previously we reported the establishment in our laboratory of a novel transwell insert culture dish method that we have used to demonstrate bystander responses, using a number of different endpoints, in AG01522 normal human fibroblasts after low LET X-irradiation (Yang et al., 2005). At the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL) we have used that transwell system, as well as irradiating cells at low fluences, to study bystander responses. We have conducted experiments every NSRL campaign since this project was funded in July 2003. To date, our studies have largely concentrated on addressing aim 1 using AG01522 fibroblasts and h-TERT immortalized human keratinocytes, although studies have been initiated on the other three aims as well. Our plan uses 1 GeV/n iron, 1 GeV/n titanium, 290 MeV/n carbon and 1 GeV/n protons because they represent particles found in space that span a range of LET values from 0.24 to about 159 keV/microm, thus allowing us to test the LET dependence of various bystander responses. With both the AG01522 fibroblasts and the keratinocytes we have found that irradiated cells show significantly increased levels of damage (loss of clonogenic survival, p21 expression and micronuclei (MN) induction) after exposure to 1 GeV/n Fe ions compared to the damage seen following 250 kVp X-rays. For example, in the AG0 cells, Relative Biological Effectiveness, RBE, values are about 2.5, 1.5 and 1.5, respectively for the three endpoints. The RBE values tend to be lower for Ti ions, and lower still for Carbon ions and protons. These data are consistent with the general dependence of RBE on LET that has been seen by others. Using our transwell insert culture system with AG01522 fibroblasts, bystander responses for production of MN (2-3-fold increase over background at doses as low as 10 cGy), loss of clonogenicity (decrease to about 80% survival) and induction of gamma-H2AX foci (~ 4-fold increase) after Fe irradiation have been found. The responses are similar in magnitude to those we have described previously with X-rays. The increased MN damage in bystander cells is independent of dose to the irradiated cells between 0.1 and 2 Gy Fe ions, although the response may be decreased slightly at 0.05 Gy. Surprisingly, to date, our data show the magnitude of the MN bystander response is smaller after proton irradiation, and no bystander response has been seen after irradiation with Ti or C ions. Studies on the time course of the bystander response suggest that the increase in gamma-H2AX foci is fully expressed within one hour after irradiation and immediate co-culturing of irradiated and unirradiated cells. Additionally, our studies show that the signaling for formation of bystander MN decays over 3-6 hours if the addition of unirradiated cells to irradiated cells is delayed. In h-TERT immortalized human keratinocytes we have also found substantial X-ray–induced bystander effects with MN as the endpoint, but there is little, if any bystander MN production after Fe-ion exposure. Furthermore, the Fe-induced increased p21 expression, as measured by immunofluorescence, in bystander cells is reduced (< 2-fold increase) compared to the response after X-ray irradiation (~ 2.5-fold). Also, the p21 expression is delayed substantially after Fe, such that after Fe ion irradiation, p21 expression in bystanders is not seen until 40-50 hours after irradiation, whereas the expression is maximal by 24 hours after X-rays. In addition to using the transwell insert system to study ion-induced bystander responses, we wanted to develop techniques to irradiate cells at very low fluences of particles, such that only 1-2% of the cells would be traversed by particles, but we could study effects in the whole population, which would include cells traversed by particles and bystander cells. Thus, if more than the 1-2% of cells irradiated showed damage, it would suggest bystander responses were operational, analogous to the method used by the Little group with alpha-particles (Nagasawa and Little, 1992). To do these experiments it is necessary to have particle fluences as low as 2000 particles/sq cm for our irradiations at NSRL. This has necessitated that Dr. Rusek and his team of NSRL physicists develop procedures for obtaining the very low fluences, and, most importantly, being able to accurately quantify the particle fluences, determine the beam uniformity and determine the purity of the beam. We have been instrumental in working with the NSRL physicists in this process. It is only in recent runs that we feel the particle fluences have been known accurately, and the data from those experiments have not yet been analyzed. However, it should be stressed that this development of the capacity for working at very low

	particle fluences is a critical advance at NSRL, not only for us but for other future users, as these fluences are now approaching the very low levels that will actually be experienced by astronauts in space. With regard to aim 2, we have found that ROS play a role in transmission of signal(s) from irradiated to bystander cells after Fe exposure, as we have shown previously with X-rays. Addition of catalase, SOD or PTIO, which scavenge hydrogen peroxide, superoxide and nitric oxide, respectively, to the medium shared by irradiated and unirradiated fibroblasts after irradiation decreases the Fe-induced bystander micronuclei formation in the fibroblasts. We have begun studies, relevant to our aim 3, using “mixed”, actually sequential, beams of protons and Fe ions. Since only one experiment has been analyzed to date, it is premature to state any conclusions. We have also begun to make progress on aim 4 in the past year. Our studies with mixed cell types in the transwell insert co-culture system show significant signaling from X-irradiated fibroblasts to unirradiated keratinocytes and vice versa (endpoints of MN formation and p21 expression). However, interestingly, after Fe exposure our preliminary data suggest that with some endpoints there is no signaling from irradiated keratinocytes to unirradiated fibroblasts or keratinocytes, and the effect is delayed from irradiated fibroblasts to unirradiated keratinocytes (maximum response in terms of MN production at 50 h rather than 24 h, as seen with X-rays). These results remain to be replicated, and mechanisms will be investigated. In summary, in the past year we have made substantial progress on all four of our original aims for this project and have made a number of interesting, novel findings in both directly irradiated and bystander cells. We plan to continue these studies in the up-coming year.
Bibliography Type:	Description: (Last Updated: 11/13/2018)
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